

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
6	(a)	(i)		phytomining uses lower grade ores than bioleaching ☐ bioleaching conserves supplies of the lowest grade metal ores ☐ phytomining conserves supplies of high-grade metal ores ☒ bioleaching uses high-grade metal ores ☐ phytomining uses the lowest grade metal ores ☐			1	1		
		(ii)		yes – plants and bacteria are both living things / living organisms must give reason			1	1		
		(iii)		award (1) for any of following - insufficient land (available to grow plants) - land is used for other things e.g. food crops - unsuitable climate (to grow plants) e.g. not sunny enough / lack of water / too hot accept any sensible answer			1	1		
		(iv)		award (1) for either of following <u>bacteria</u> extract (the copper) at a low rate <u>bacterial</u> growth is slow / have to wait for the bacteria to grow			1	1		